Data Visual

ToyCraft Tales: Tableau's Vision into Toy Manufacturer Data

1. Preprocessing Steps

Data preprocessing was performed in two stages: SQL level and Tableau level to ensure accuracy and reliability of the analysis.

1.1 Data Cleaning using SQL

The following preprocessing steps were performed in SQL:

- Removed duplicate records from manufacturer dataset.
- Handled NULL values using appropriate filtering and replacement.
- Standardized state names to maintain consistency.
- Converted numerical fields to proper numeric data types.
- Formatted year column correctly for time-series analysis.
- Verified data integrity using SELECT and COUNT queries.
- Used GROUP BY to calculate:
    - State-wise manufacturer count
    - Year-wise manufacturer trend
- Sorted data using ORDER BY for ranking (Top 10 states).
- Ensured index values were correctly mapped to respective states.

These steps ensured the dataset was clean, structured, and analysis-ready.

1.2 Data Preparation in Tableau

After importing the cleaned dataset into Tableau:

- Verified correct data types (String, Number, Date).
- Created calculated fields where required.
- Applied sorting to display Top 10 states.
- Created hierarchies if needed (Year → State).
- Used filters for better interactive analysis.
- Designed dashboard layout for clarity and readability.

This improved visualization clarity and performance.

2 Business Questions with Visualization

The following business questions were identified and answered using Tableau visualizations:

2.1 Which are the Top 10 states with highest toy manufacturers?

- Visualization Used: Horizontal Bar Chart
  Purpose:
  To compare manufacturer count across states and identify leading states.

- Insight:
  Some states have significantly higher manufacturing presence compared to others.

2.2 How has the number of toy manufacturers changed over the years?

- Visualization Used: Line Chart (Trend Analysis)
  Purpose:
  To analyze growth or decline trends across different years.

- Insight:
  The industry shows fluctuations over time, indicating possible economic or market influence.

2.3 What is the relationship between Index and Manufacturer Count?

- Visualization Used: Dual-axis Bar Chart
  Purpose:
  To analyze how index values relate to manufacturer concentration.

- Insight:
  States with higher index values generally show stronger manufacturing performance.

2.4 What is the overall distribution of toy manufacturers?

- Visualization Used: Vertical Bar Chart
  Purpose:
  To understand total manufacturer distribution.

- Insight:
  The data highlights overall manufacturer strength and comparative performance.

2.5 What percentage of toy manufacturers are distributed across US states?

- Visualization Used: Pie Chart
  Purpose:
  To show percentage contribution of each state.

- Insight:
  Few states dominate the overall share of toy manufacturing.

Conclusion

Through structured preprocessing in SQL and visualization in Tableau, the Toy Manufacturing dataset was transformed into meaningful business insights.

The project successfully identified:

- Leading manufacturing states
- Year-wise industry trends
- Index-based performance
- Percentage distribution across states

This structured approach ensures reliable analysis and supports strategic decision-making.